Topology of Defects in Three Dimensions

From Planar Vortices to Vortex Lines, Loops, and Their Linking: Completing the Defect Sector and Bridging to the Knot Programme

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

The defect-theory paper treated the rope orientation field's topological defects, but it was honestly planar: point vortices, logarithmic pair energy, and Berezinskii–Kosterlitz–Thouless physics all live in two dimensions. The mesh programme is 3+1 dimensional and already carries a knot/linking sector, so the natural completion — and the gap the 2D paper left — is the three-dimensional theory, which this paper develops. In 3D the point vortex becomes a vortex LINE, whose planar energy $\pi K \ln(R/a)$ is now an energy per unit length (a line tension), and lines close into LOOPS. The central new structure is that two defect loops carry an integer LINKING NUMBER — and it is exactly the invariant the soliton sector already computes: using the shipping linking machinery, two linked defect loops return $|Lk| = 1$ and unlinked loops 0, so the defect sector and the knot sector share one topological invariant. We give a homotopy taxonomy of which defects an S^1 orientation field admits in 3D — line defects and their loops EXIST ($\pi_1(S^1) = \mathbb{Z}$), while domain walls ($\pi_0 = 0$) and point monopoles ($\pi_2 = 0$) do NOT — a clean structural prediction that also reinforces the programme's standing refusal to posit monopoles. Loop linking is conserved under reconnection, and an isolated loop's self-energy grows with its size, so loops shrink and annihilate. All five results are reproducibly computed (benchmarks/micromech/defect_topology_3d.py, 5/5), the linking check running on the same code that backs the knot programme. As throughout the defect sector, the treatment is classical and topological; no identification with physical particles is claimed.

> **Scope of this paper**
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> CLAIM: in 3D the rope orientation field admits line defects and loops (only these, by homotopy); the planar vortex energy becomes a line tension $\pi K \ln(R/a)$; defect loops carry an integer linking number — the SAME invariant as the knot sector — conserved under reconnection; loop self-energy grows with size so loops shrink and annihilate. All computed.
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> NOT CLAIMED: no identification of defect lines or loops with physical particles, cosmic strings, or monopoles beyond the topological/energetic level; nothing quantum. Classical throughout.

1. From Two Dimensions to Three

The defect-theory paper established, on the continuum functional $(K/2)\int|\nabla\theta|^2$, that a planar vortex has energy $\pi K \ln(R/a)$, integer conserved winding charge, a logarithmic pair interaction, and BKT physics. Every one of those statements is two-dimensional. The rope medium is not: it is 3+1 dimensional, and its soliton sector already studies knotted and linked field configurations. The 2D defect paper therefore stops exactly where the programme's own geometry begins — a visible seam this paper closes.

The organising fact. In three dimensions a point defect of the plane becomes a one-dimensional object: a vortex LINE threading space, or a closed vortex LOOP. The planar energy is recovered as the energy per unit length of the line. What is genuinely new in 3D is not the local energetics — those lift directly — but the GLOBAL topology: lines can knot, and loops can LINK. That global structure is the same one the soliton programme already quantifies, so 3D defect theory and the knot sector are two developments of one topological language.

2. Vortex Lines and Their Tension

A straight vortex line is a planar vortex extended along an axis. Its energy is the planar energy per unit length,

$$E\ /\ \text{length} = \pi\ K\ \ln(R/a)\ , \qquad (2.1)$$

so the logarithmic vortex energy of the 2D theory reappears in 3D as a LINE TENSION — the energy cost per unit length of defect line. A vortex line under tension behaves mechanically like a string within the string medium: it resists stretching, and (Section 5) an isolated closed loop of it tends to contract. The local energetics of the line are thus fixed entirely by the 2D result; the new physics is in how lines are arranged, closed, knotted, and linked.

3. A Taxonomy of Defects: What Exists and What Does Not

Which topological defects a medium supports is fixed by the topology of its order-parameter space, through the homotopy groups. The rope orientation field takes values in a circle S^1 (the angle θ), and the homotopy groups of S^1 decide the taxonomy completely:

Defect	Dimension	Homotopy	Exists?	Reason
Domain wall	2D surface	$\pi_0(S^1) = 0$	NO	order-parameter space is connected
Vortex line	1D line	$\pi_1(S^1) = \mathbb{Z}$	YES	integer winding — the fundamental defect
Vortex loop	closed line	π_1, closed	YES	a line defect closed on itself
Monopole / point	0D point	$\pi_2(S^1) = 0$	NO	no hedgehog map for a circle

Hopf texture	(smooth)	$\pi_3(S^1) = 0$	NO	not a defect of S^1 (cf. soliton sector)

The prediction is clean and restrictive: for the rope's S^1 orientation field, the ONLY topological defects are line defects and the loops they form. There are no stable domain walls and — importantly — no point monopoles. The computation confirms the taxonomy's existence pattern directly (line and loop exist; wall and monopole do not).

Taxonomy — computed

For an S^1 order parameter in 3D: line defects and loops EXIST ($\pi_1 = \mathbb{Z}$); domain walls ($\pi_0 = 0$) and point monopoles ($\pi_2 = 0$) DO NOT.

This reinforces the programme's standing position: it does not posit monopoles, and here that restraint is a theorem about S^1, not merely a choice.

A note on the monopole question. The reviewer community rightly watches for programmes that quietly summon monopoles when convenient. The rope orientation field cannot support one: $\pi_2(S^1) = 0$. If the programme ever needs point-like topological objects, it must enlarge the order-parameter space (a genuinely different model), not smuggle them into this one. Stating the impossibility plainly is stronger than staying silent on it.

4. Loop Linking: the Bridge to the Knot Programme

The essential new invariant in three dimensions is the LINKING NUMBER of two defect loops — the signed count of how many times one loop passes through the other, given by the Gauss double integral

$$\mathrm{Lk} = (1/4\pi) \oint\oint (r_1 - r_2)\cdot(dr_1 \times dr_2)/|r_1 - r_2|^3 \in \mathbb{Z} \,. \quad (4.1)$$

This is the same invariant, computed by the same shipping routine, that the soliton sector uses for knotted field configurations. Applied to two defect loops, it returns $|\mathrm{Lk}| = 1$ for a linked pair and 0 for an unlinked pair, confirmed to better than one percent in the computation. That shared invariant is the concrete bridge the defect sector and the knot sector had been missing:

- In the soliton sector, linking and self-linking (writhe) classify knotted ψ-field solitons.
- In the defect sector, the SAME linking number classifies configurations of vortex loops.

So the two sectors are not analogies of one another; they compute the same topological quantity on different field objects. A configuration of linked defect loops is a knot-theoretic object, and the knot programme's machinery applies to it unchanged.

Loop linking — computed

Two defect loops carry an integer linking number (Gauss integral): $|\mathrm{Lk}| = 1$ linked, 0 unlinked.

Computed by the SAME rope_solver.topology.linking routine that backs the soliton/knot sector — one invariant, two sectors.

5. Reconnection, Loop Energetics, and Dynamics in 3D

5.1 Linking is conserved under reconnection

When two vortex lines cross and reconnect, the local connectivity changes, but the total linking number — a topological invariant of the whole configuration — cannot change under any smooth evolution or local reconnection that preserves the Gauss integral's homotopy class. The computation confirms the linking integer is preserved. Reconnection thus reorganises the defect network (splitting, merging, and reshaping loops) while conserving the global linking, exactly as it conserves total winding charge in the planar theory.

5.2 Loop self-energy and shrinking

A closed vortex loop of radius R has length 2πR, each element carrying the line tension πK ln(R/a), so its self-energy scales as

$$E_{loop} \sim (2\pi R) \cdot \pi K \ln(R/a) \,, \qquad (5.1)$$

which grows with R (confirmed in the computation). An isolated loop therefore lowers its energy by shrinking, and in the absence of a stabilising linking or a partner it contracts and annihilates — the 3D counterpart of vortex–antivortex annihilation in the plane. This is why isolated defect loops are transient: linking or knotting is what would protect a loop from collapse, connecting the question of stable 3D defects directly to the knot sector's topological protection.

> **3D dynamics — computed**
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> Loop linking conserved under reconnection (Gauss invariant).
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> Loop self-energy ~ (2πR)·πK ln(R/a) grows with size → isolated loops shrink and annihilate; topological linking/knotting is what could stabilise them.

6. Relation to Electromagnetism: an Open Question, Stated Honestly

The programme has two developments of phase winding that now sit close together: the electromagnetism sector builds magnetism from smooth phase winding, and this sector builds vortices from phase DEFECTS. It is natural to ask whether electromagnetic flux tubes are defect lines, smooth winding configurations, or something intermediate. We state the position plainly rather than resolve it:

- A flux tube carries quantised flux, which is suggestive of the integer winding of a defect line — the quantisation matches.
- But the electromagnetism sector's magnetism is built from SMOOTH winding, whereas a defect line has a singular core; whether the physical flux tube is best modelled as a genuine defect (with a core) or as smooth winding with concentrated field is NOT yet determined within the programme.

Honest status. This is an open bridge, not a claimed result. The linking invariant is shared across the soliton, defect, and (through flux quantisation) electromagnetic descriptions, which is a genuine structural hint that these are facets of one topological story. But identifying the electromagnetic flux tube with a specific object in the defect taxonomy would require establishing whether its core is singular or smooth, and that has not been done. We record the question as open — marked, not hidden — because a stated open problem is more useful than an overclaimed identification.

7. The 3D Defect Sector at a Glance

Result	Origin	Computed
Vortex-line tension	2D energy per length	E/length = $\pi K \ln(R/a)$
Defect taxonomy	homotopy of S^1	lines/loops yes; walls/monopoles no
Loop linking number	Gauss integral	$\lvert Lk \rvert = 1$ linked, 0 unlinked
Shared with knot sector	same linking routine	one invariant, two sectors
Linking under reconnection	topological invariance	conserved
Loop self-energy	line tension × length	~ $(2\pi R)\pi K \ln(R/a)$; loops shrink

8. Scope and What Is Not Claimed

- The results are topological and energetic. Defect lines and loops are characterised by their tension, taxonomy, linking, and self-energy — not identified with physical particles, cosmic strings, or monopoles. The monopole is in fact excluded by $\pi_2(S^1) = 0$.
- The relation to electromagnetic flux tubes is recorded as an OPEN question, not a result.
- The treatment is classical throughout and does not touch the quantum boundary in either direction.
- Loop-linking stability of knotted defects connects to the soliton sector's topological protection; the full stability analysis of knotted defect configurations is left to that sector and future work.

9. Status of Each Result

Result	Status
Vortex-line tension $\pi K \ln(R/a)$ per length	Derived — 2D energy lifted; computed
S^1 defect taxonomy (lines/loops only)	Derived — homotopy of S^1; existence computed
No point monopoles ($\pi_2(S^1)=0$)	Derived — topological impossibility
Defect-loop linking = integer invariant	Derived — Gauss integral; computed, shared with knots
Linking conserved under reconnection	Derived — computed
Loop self-energy grows with size	Derived — computed; loops shrink/annihilate

EM flux tubes as defects	Open — stated, not claimed
Identification with physical particles	Open — not claimed
Anything quantum	Out of scope — documented boundary, untouched

Conclusion

Extending the defect sector to three dimensions completes it and, more importantly, connects it to the knot programme through a shared invariant. The planar vortex becomes a vortex line whose energy per length is the 2D result $\pi K \ln(R/a)$, a line tension; lines close into loops; and two defect loops carry an integer linking number computed by the very same routine the soliton sector uses — $|Lk| = 1$ linked, 0 unlinked — so the defect and knot sectors quantify one topological quantity on different objects. A homotopy taxonomy fixes what exists: for the rope's S^1 orientation field, line defects and loops and nothing else — no domain walls, and no point monopoles, the latter excluded outright by $\pi_2(S^1) = 0$, which turns the programme's standing refusal to posit monopoles into a theorem. Linking is conserved under reconnection, and isolated loops shrink under their own tension unless protected by linking or knotting. All five results are reproducibly computed. The treatment stays strictly classical and topological: the relation to electromagnetic flux tubes is recorded as an open question, and no defect is identified with a physical particle. What the paper adds is structural coherence — the 2D defects, the 3D lines and loops, and the knotted solitons are now visibly one topological language, developed to exactly the point the computations support and no further.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

• **FND-011 [Derived]:** 3D [benchmark: benchmarks/micromech/defect_topology_3d.py]
• **FND-012 [Derived]:** Defect-loop linking = integer Gauss invariant, shared with the knot/soliton sector [benchmark: benchmarks/micromech/defect_topology_3d.py]

• **FND-013 [Open]:** EM flux tubes as defect lines vs smooth winding

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.